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CS 2300 : Discrete Computational Structures

Summer Semester, 2025

HOMEWORK ASSIGNMENT #4

Due Date: Friday, June 20

These are the problems that you need to hand in for grading. Always explain your answers and show your reasoning. **The total points add up to 100, but you will be graded out of 90. This means you have the opportunity to earn 10 extra credit points.**

1 Induction

1. [6 Pts] $1^3 + 2^3 + \cdots + n^3 = (n(n+1)/2)^2$, for all positive integers n .

Base Case: Let $n = 1$.

Left side: $1^3 = 1$

Right side: $\left(\frac{1(1+1)}{2}\right)^2 = \left(\frac{2}{2}\right)^2 = 1$

Since both sides are equal, the base case holds.

Inductive Hypothesis:

Assume the formula holds for $n = k$:

$$1^3 + 2^3 + \cdots + k^3 = \left(\frac{k(k+1)}{2}\right)^2$$

Inductive Step:

Show that the formula holds for $k + 1$:

$$1^3 + 2^3 + \cdots + k^3 + (k+1)^3 = \left(\frac{(k+1)(k+2)}{2}\right)^2$$

Use inductive hypothesis:

$$1^3 + 2^3 + \cdots + k^3 + (k+1)^3 = \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3$$

Factor out $(k+1)^2$:

$$= (k+1)^2 \left(\left(\frac{k}{2}\right)^2 + (k+1) \right) = (k+1)^2 \left(\frac{k^2}{4} + (k+1) \right)$$

$$\text{Simplify: } \frac{k^2}{4} + (k+1) = \frac{k^2+4k+4}{4} = \frac{(k+2)^2}{4}$$

$$\text{Therefore: } (k+1)^2 \cdot \frac{(k+2)^2}{4} = \left(\frac{(k+1)(k+2)}{2}\right)^2 \text{ which holds.}$$

The formula holds for all positive integers n by mathematical induction.

2. [6 Pts] $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n + 1)! - 1$, for all positive integers n .

Base Case: Let $n = 1$.

Left side: $1 \cdot 1! = 1$

Right side: $(1 + 1)! - 1 = 2! - 1 = 2 - 1 = 1$

Since both sides are equal, the base case holds.

Inductive Hypothesis:

Assume the formula holds for $n = k$:

$$1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! = (k + 1)! - 1$$

Inductive Step:

Show that the formula holds for $k + 1$:

$$1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! + (k + 1)(k + 1)! = (k + 2)! - 1$$

Use the inductive hypothesis:

$$1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! + (k + 1)(k + 1)! = (k + 1)! - 1 + (k + 1)(k + 1)!$$

Factor out $(k + 1)!$:

$$= (k + 1)!(1 + (k + 1)) - 1 = (k + 1)! \cdot (k + 2) - 1$$

Simplify:

$$(k + 1)! \cdot (k + 2) = (k + 2)!, \text{ so I have:}$$

$$(k + 2)! - 1$$

Which matches the right side.

$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n + 1)! - 1$ holds for all positive integers n by mathematical induction.

3. [6 Pts] $2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^n = (1 - (-7)^{n+1})/4$, for all non-negative integers n .

Base Case: Let $n = 0$.

Left side: $2(-7)^0 = 2 \cdot 1 = 2$

Right side: $\frac{1 - (-7)^{0+1}}{4} = \frac{1 - (-7)}{4} = \frac{8}{4} = 2$

Since both sides are equal, the base case holds.

Inductive Hypothesis:

Assume the formula holds for $n = k$:

$$2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^k = \frac{1 - (-7)^{k+1}}{4}$$

Inductive Step:

Show that the formula holds for $k + 1$:

$$2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^k + 2(-7)^{k+1} = \frac{1 - (-7)^{k+2}}{4}$$

Use the inductive hypothesis:

$$\text{Left side} = \left(\frac{1 - (-7)^{k+1}}{4} \right) + 2(-7)^{k+1} = \frac{1 - (-7)^{k+1} + 8(-7)^{k+1}}{4} = \frac{1 + 7(-7)^{k+1}}{4}$$

$$\text{Also, } 1 + 7(-7)^{k+1} = 1 - (-7)^{k+2}$$

$$\text{So, } \frac{1 + 7(-7)^{k+1}}{4} = \frac{1 - (-7)^{k+2}}{4}$$

Which matches the right-hand side.

The formula holds for all non-negative integers n by mathematical induction.

4. [6 Pts] $1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} < 3 - \frac{1}{n}$, for all integers $n \geq 2$.

Base Case: Let $n = 2$.

Left side: $1 + \frac{1}{4} = \frac{5}{4} = 1.25$

Right side: $3 - \frac{1}{2} = \frac{5}{2} = 2.5$

Since $1.25 < 2.5$, the base case holds.

Inductive Hypothesis:

Assume the inequality holds for $n = k$, where $k \geq 2$:

$$1 + \frac{1}{4} + \cdots + \frac{1}{k^2} < 3 - \frac{1}{k}$$

Inductive Step:

Show it holds for $k + 1$:

$$1 + \frac{1}{4} + \cdots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 3 - \frac{1}{k+1}$$

Start with the left side and apply the inductive hypothesis:

$$1 + \frac{1}{4} + \cdots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < \left(3 - \frac{1}{k}\right) + \frac{1}{(k+1)^2}$$

I want to show:

$$\left(3 - \frac{1}{k}\right) + \frac{1}{(k+1)^2} < 3 - \frac{1}{k+1}$$

Subtract 3 from both sides:

$$-\frac{1}{k} + \frac{1}{(k+1)^2} < -\frac{1}{k+1}$$

Add $\frac{1}{k}$ to both sides:

$$\frac{1}{(k+1)^2} < \frac{1}{k} - \frac{1}{k+1}$$

Now simplify the right-hand side:

$$\frac{1}{k} - \frac{1}{k+1} = \frac{(k+1)-k}{k(k+1)} = \frac{1}{k(k+1)}$$

So it suffices to show:

$$\frac{1}{(k+1)^2} < \frac{1}{k(k+1)}$$

Multiply both sides by $(k+1)^2 k(k+1)$ (positive for $k \geq 2$):

$$k < k+1 \quad \text{which is true}$$

The inequality holds for $k + 1$, so by the principle of mathematical induction, the inequality $1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} < 3 - \frac{1}{n}$ holds for all integers $n \geq 2$.

5. [6 Pts] 3 divides $n^3 + 2n$, for all positive integers n .

Base Case: Let $n = 1$.

Left side: $1^3 + 2 \cdot 1 = 1 + 2 = 3$

Since 3 divides 3, the base case holds.

Inductive Hypothesis:

Assume 3 divides $(k^3 + 2k)$ for some integer $k \geq 1$. That is,

$$k^3 + 2k \equiv 0$$

Inductive Step:

I want to show that 3 divides $((k + 1)^3 + 2(k + 1))$.

Compute:

$$\begin{aligned}(k + 1)^3 + 2(k + 1) &= k^3 + 3k^2 + 3k + 1 + 2k + 2 \\ &= k^3 + 3k^2 + 5k + 3\end{aligned}$$

Group terms:

$$= (k^3 + 2k) + 3k^2 + 3k + 3$$

By the inductive hypothesis, $k^3 + 2k \equiv 0 \pmod{3}$, and the rest of the terms are clearly divisible by 3.

Therefore:

$$(k + 1)^3 + 2(k + 1) \equiv 0$$

By the principle of mathematical induction, 3 divides $n^3 + 2n$ for all positive integers n .

6. [6 Pts] $3^n < n!$ for all integers $n \geq 7$. In addition, show that the statement is false for all positive integers $n < 7$.

Check for $n < 7$:

$$n = 1 : \quad 3^1 = 3, \quad 1! = 1 \quad \Rightarrow 3 > 1$$

$$n = 2 : \quad 3^2 = 9, \quad 2! = 2 \quad \Rightarrow 9 > 2$$

$$n = 3 : \quad 3^3 = 27, \quad 3! = 6 \quad \Rightarrow 27 > 6$$

$$n = 4 : \quad 3^4 = 81, \quad 4! = 24 \quad \Rightarrow 81 > 24$$

$$n = 5 : \quad 3^5 = 243, \quad 5! = 120 \quad \Rightarrow 243 > 120$$

$$n = 6 : \quad 3^6 = 729, \quad 6! = 720 \quad \Rightarrow 729 > 720$$

So the statement is false for all positive integers $n < 7$.

Base Case: Let $n = 7$.

$$3^7 = 2187, \quad 7! = 5040, \quad \text{and} \quad 2187 < 5040$$

So the base case holds.

Inductive Hypothesis:

Assume $3^k < k!$ for some integer $k \geq 7$.

Inductive Step:

I want to show that $3^{k+1} < (k+1)!$.

I start with:

$$3^{k+1} = 3 \cdot 3^k$$

By the inductive hypothesis, $3^k < k!$, so:

$$3^{k+1} < 3 \cdot k!$$

It remains to show that $3 \cdot k! < (k+1)! = (k+1) \cdot k!$

This simplifies to:

$$3 < k+1$$

Which is true for all $k \geq 7$, since $k+1 \geq 8$.

Therefore:

$$3^{k+1} < 3 \cdot k! < (k+1) \cdot k! = (k+1)!$$

By the principle of mathematical induction, $3^n < n!$ holds for all integers $n \geq 7$, and the statement is false for all positive integers $n < 7$.

7. [12 points] Let $P(n)$ be the statement that n -cent postage can be formed using just 4-cent and 7-cent stamps. Prove that $P(n)$ is true for all $n \geq 18$, using the steps below.

- (a) First, prove $P(n)$ by **regular induction**. State your basis step and inductive step clearly and prove them.

Base Cases:

Check the base cases $n = 18, 19, 20, 21$:

$$P(18) : \quad 7 + 7 + 4 = 18$$

$$P(19) : \quad 7 + 4 + 4 + 4 = 19$$

$$P(20) : \quad 4 + 4 + 4 + 4 + 4 = 20$$

$$P(21) : \quad 7 + 7 + 7 = 21$$

Thus, $P(18), P(19), P(20), P(21)$ are all true.

Inductive Step:

Assume: For some $k \geq 21$, the statement $P(k)$ is true.

Prove that $P(k + 1)$ is true.

Since $k \geq 21$, I have $k - 3 \geq 18$. By the inductive hypothesis, $P(k - 3)$ is true, so I can form postage of $k - 3$ cents.

Then, by adding one 4-cent stamp to the combination that forms $k - 3$, I obtained postage of $k + 1$ cents:

$$P(k - 3) \Rightarrow P(k + 1)$$

Therefore, $P(k + 1)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all integers $n \geq 18$. Hence, any amount of postage $n \geq 18$ can be formed using only 4-cent and 7-cent stamps.

- (b) Now, prove $P(n)$ by **strong induction**. Again, state and prove your basis step and inductive step. Your basis step should have multiple cases.

Basis Step

Verify $P(n)$ for the initial values $n = 18, 19, 20, 21$:

$$P(18) : \quad 7 + 7 + 4 = 18$$

$$P(19) : \quad 7 + 4 + 4 + 4 = 19$$

$$P(20) : \quad 4 + 4 + 4 + 4 + 4 = 20$$

$$P(21) : \quad 7 + 7 + 7 = 21$$

Thus, $P(18), P(19), P(20), P(21)$ are all true.

Inductive Step

Assume: For some $k \geq 21$, the statements $P(m)$ are true for all integers m with $18 \leq m \leq k$.

Show that $P(k + 1)$ is also true.

Since $k - 3 \geq 18$, by the inductive hypothesis, I know that $P(k - 3)$ is true. I can form postage of $k - 3$ cents using 4-cent and 7-cent stamps.

Then, adding one 4-cent stamp gives:

$$(k - 3) + 4 = k + 1$$

So I can form postage of $k + 1$ cents.

Therefore, $P(k + 1)$ is true.

By the principle of strong induction, $P(n)$ is true for all integers $n \geq 18$. Hence, any amount of postage $n \geq 18$ can be formed using only 4-cent and 7-cent stamps.

8. [6 points] Suppose $P(n)$ is true for an infinite number of positive integers n . Also, suppose that $P(k+1) \rightarrow P(k)$ for all positive integers k . Now, prove that $P(n)$ is true for all positive integers. This is the *reverse induction principle*.

Hint: Use proof by contradiction

Proof by Contradiction

Suppose that $P(n)$ is not true for all positive integers n . That is, define the set

$$S = \{n \in \mathbb{Z}^+ : P(n) \text{ is false}\}$$

Assume $S \neq \emptyset$.

Since S is a non-empty subset of the positive integers, it must have a least element by the Well-Ordering Principle.

Let m be the least element of S . Then $P(m)$ is false, but $P(n)$ is true for all $n > m$.

In particular, $P(m+1)$ must be true, since $m+1 > m$ and m is the smallest counterexample.

But I am given that $P(k+1) \rightarrow P(k)$ for all $k \in \mathbb{Z}^+$.

Applying this to $k = m$, I have $P(m+1) \rightarrow P(m)$.

Since $P(m+1)$ is true, it follows that $P(m)$ is also true. This contradicts the assumption that $P(m)$ is false.

Therefore, the assumption that $S \neq \emptyset$ must be incorrect. Hence, $S = \emptyset$ and $P(n)$ is true for all positive integers n .

9. **[6 points]** Prove that $f_0 - f_1 + f_2 - \cdots - f_{2n-1} + f_{2n} = f_{2n-1} - 1$ where f_i are the Fibonacci numbers.

For all positive integers n ,

$$f_0 - f_1 + f_2 - f_3 + \cdots - f_{2n-1} + f_{2n} = f_{2n-1} - 1,$$

where f_i denotes the i -th Fibonacci number, defined by:

$$f_0 = 0, \quad f_1 = 1, \quad f_n = f_{n-1} + f_{n-2} \text{ for } n \geq 2$$

Base Case: Let $n = 1$.

$$\text{Left Side: } f_0 - f_1 + f_2 = 0 - 1 + 1 = 0$$

$$\text{Right Side: } f_{2n-1} - 1 = f_1 - 1 = 1 - 1 = 0$$

Thus, the identity holds for $n = 1$.

Inductive Step: Assume the identity holds for some $n = k$; that is,

$$f_0 - f_1 + f_2 - f_3 + \cdots - f_{2k-1} + f_{2k} = f_{2k-1} - 1$$

I want to prove that:

$$f_0 - f_1 + f_2 - f_3 + \cdots - f_{2k+1} + f_{2k+2} = f_{2k+1} - 1$$

Let S_{k+1} denote the left-hand side up to f_{2k+2} . Then:

$$S_{k+1} = (f_0 - f_1 + \cdots - f_{2k-1} + f_{2k}) - f_{2k+1} + f_{2k+2}$$

By the inductive hypothesis, the expression in parentheses is $f_{2k-1} - 1$, so:

$$S_{k+1} = f_{2k-1} - 1 - f_{2k+1} + f_{2k+2}$$

The Fibonacci recurrence relation:

$$f_{2k+2} = f_{2k+1} + f_{2k}$$

Substitute:

$$S_{k+1} = f_{2k-1} - 1 - f_{2k+1} + (f_{2k+1} + f_{2k}) = f_{2k-1} - 1 + f_{2k}.$$

Use identity:

$$f_{2k} + f_{2k-1} = f_{2k+1}$$

so:

$$S_{k+1} = f_{2k+1} - 1.$$

Thus, the identity holds for $n = k + 1$. By the principle of mathematical induction, the identity holds.

10. [8 points] Let $P(n)$ be the statement that n can be written as a sum of distinct powers of 2. For example, $5 = 2^0 + 2^2$ and $10 = 2^1 + 2^3$. Note that $10 = 2^0 + 2^0 + 2^3$ is not a sum of *distinct* powers of 2. Prove that $P(n)$ is true for all positive integers n by strong induction.

Claim $P(n)$: For every positive integer n , n can be written as a sum of *distinct* powers of 2.

Proof by Strong Induction:

Base Case: Let $n = 1$.

Then $1 = 2^0$ which is a sum of distinct powers of 2. Thus, $P(1)$ is true.

Inductive Hypothesis: Assume: For all integers k such that $1 \leq k \leq n$, $P(k)$ is true;

Inductive Step:

I want to show that $P(n + 1)$ is true.

Let: $m = \lfloor \log_2(n + 1) \rfloor$

so that

$$2^m \leq n + 1 < 2^{m+1}$$

Then I can write

$$n + 1 = 2^m + r$$

for some r with

$$0 \leq r < 2^m$$

Since $r < 2^m$ and $r < n + 1$, I have $r \leq n$.

By the inductive hypothesis, r can be expressed as a sum of distinct powers of 2.

Because 2^m is strictly greater than any power of 2 in the sum for r , the powers of 2 in the sums for 2^m and r are distinct.

Hence, $n + 1 = 2^m + r$ is a sum of distinct powers of 2.

By strong induction, $P(n)$ holds for all positive integers n .

11. [12 Pts] Let S defined recursively by

- (1) Base step: $4 \in S$ and
- (2) Inductive step: if $s \in S$ and $t \in S$, then $st \in S$.

Let $A = \{4^i \mid i \in \mathbb{Z}^+\}$. Prove that

(a) [6 Pts] $A \subseteq S$ by mathematical induction.

Base Case: $n = 1$

I have $4^1 = 4 \in S$ by the base case of the recursive definition.

Therefore, $P(1)$ holds.

Inductive Hypothesis:

Assume: For some $k \in \mathbb{Z}^+$, $4^k \in S$.

Inductive Step:

We want to show that $4^{k+1} \in S$.

Since $4^k \in S$ (by the inductive hypothesis) and $4 \in S$ (by the base case), then by the inductive rule (if $s \in S$ and $t \in S$, then $st \in S$), I have:

$$4^k \cdot 4 = 4^{k+1} \in S$$

Thus, $P(k+1)$ holds.

By the principle of mathematical induction, $4^n \in S$ for all $n \in \mathbb{Z}^+$.

Therefore, $A = \{4^i \mid i \in \mathbb{Z}^+\} \subseteq S$.

(b) [6 Pts] $S \subseteq A$ by structural induction.

Base Case:

I am given that $4 \in S$.

$4 = 4^1 \in A$, so the base case holds.

Inductive Step:

Suppose $s, t \in S$, and by the inductive hypothesis, there exist positive integers $i, j \in \mathbb{Z}^+$ such that:

$$s = 4^i \quad \text{and} \quad t = 4^j$$

Then by the recursive definition, $st \in S$, and:

$$st = 4^i \cdot 4^j = 4^{i+j}$$

Since $i+j \in \mathbb{Z}^+$, I can conclude that $st \in A$.

By structural induction, every element of S belongs to A . Therefore, $S \subseteq A$.

12. [8 Pts] Inductive definition for Structural construction (No Proof required) [Give base step and inductive step]

- (a) [4 Pts] Give an inductive definition of the set of palindromes over the alphabet $\{a, b, c\}$. You do not need to prove that your construction is correct. *Note:* a , b , c , aa , cc , aba are all palindromes.

Base Step: The empty string ε and each single-character string a , b , and c are in the set of palindromes P .

That is, $\varepsilon, a, b, c \in P$.

Inductive Step: If $w \in P$ and $x \in \{a, b, c\}$, then $xwx \in P$.

- (b) [4 Pts] Define the set $S = \{2^k 3^m \mid k, m \in \mathbb{Z}^+\}$ inductively. You do not need to prove that your construction is correct.

Base Step:

$2 \cdot 3 = 6 \in S$. That is, $2^1 3^1 = 6 \in S$.

Inductive Step:

If $s \in S$, then $2s \in S$.

If $s \in S$, then $3s \in S$.

13. [12 Pts] Given the inductive definition of full binary trees (FBTs), define $n(T)$, the number of vertices in tree T , and $\ell(T)$, the number of leaves in tree T , inductively. Then, use structural induction to prove that for all FBTs T , $n(T) = 2\ell(T) - 1$.

Full Binary Tree (FBT):

Base Case: A single vertex (leaf) is a FBT.

Inductive Step: If T_1 and T_2 are FBTs, then joining them under a new root gives a FBT T .

Number of vertices $n(T)$:

Base: $n(T) = 1$

Step: $n(T) = 1 + n(T_1) + n(T_2)$

Number of leaves $\ell(T)$:

Base: $\ell(T) = 1$

Step: $\ell(T) = \ell(T_1) + \ell(T_2)$

Structural Induction Proof:

Base Case: If T is a single leaf:

$$n(T) = 1, \quad \ell(T) = 1, \quad 2\ell(T) - 1 = 1$$

So, the formula holds.

Inductive Step:

Assume it holds for T_1 and T_2 :

$$n(T_1) = 2\ell(T_1) - 1, \quad n(T_2) = 2\ell(T_2) - 1$$

$$\begin{aligned} \text{For } T \text{ formed by joining } T_1 \text{ and } T_2: & n(T) = 1 + n(T_1) + n(T_2) \\ &= 1 + (2\ell(T_1) - 1) + (2\ell(T_2) - 1) \\ &= 2(\ell(T_1) + \ell(T_2)) - 1 \\ &= 2\ell(T) - 1 \end{aligned}$$

By structural induction, for all FBTs T , I have: $n(T) = 2\ell(T) - 1$.